

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

RELATIONS & FUNCTIONS **PART-02 (71 to 130)**

DIBY (DO IT BY YOURSELF)

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DOMAIN OF FUNCTION

DIBY-71

If the domain of the function $f(x) = \frac{1}{\sqrt{10+3x-x^2}} + \frac{1}{\sqrt{x+|x|}}$ is (a, b) , then

$(1+a)^2 + b^2$ is equal to:

[JEE Mains-2025]

[Ans. : 26]

DIBY-72

If the domain of the function $f(x) = \sin^{-1} \left(\frac{1}{x^2-2x-2} \right)$ is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then

$\alpha + \beta + \gamma + \delta$ is equal to

[JEE Mains-2026]

[Ans. : 4]

DIBY-73

The domain of the function $f(x) = \sin^{-1} \left(\frac{|x|+5}{x^2+1} \right)$ is $(-\infty, -a] \cup [a, \infty)$. Then a is equal to

[JEE Mains-2020]

(A) $\frac{\sqrt{17}-1}{2}$

(B) $\frac{\sqrt{17}}{2}$

(C) $\frac{\sqrt{17}}{2} + 1$

(D) $\frac{1+\sqrt{17}}{2}$

[Ans. : (D)]

QUESTIONS ON LOG

DIBY-74

If $x = \log_a(bc)$, $y = \log_b(ca)$ and $z = \log_c(ab)$, then the value of $(1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1} = \underline{\hspace{2cm}}$.

[Ans. : 1]

DIBY-75

Suppose $x, y, z > 0$ and different than one and $\ln x + \ln y + \ln z = 0$.

If $\frac{1}{e^k} = x^{\frac{1}{\ln y} + \frac{1}{\ln z}} \cdot y^{\frac{1}{\ln z} + \frac{1}{\ln x}} \cdot z^{\frac{1}{\ln x} + \frac{1}{\ln y}}$. The $k =$

[Ans. : 3]

DIBY-76

$$\log_{\frac{3}{4}} \log_8(x^2 + 7) + \log_{\frac{1}{2}} \log_{\frac{1}{4}}(x^2 + 7)^{-1} = -2$$

[Ans. : $x = 3$ or -3]

DIBY-77

The product of all positive real values of x satisfying the equation

$$x^{(16(\log_5 x)^3 - 68\log_5 x)} = 5^{-16} \text{ is}$$

[JEE (Adv.)-2022]

[Ans.: 1]

DIBY-78

Let (x_0, y_0) be the solution of the following equations

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is

[JEE (Adv.)-2011]

(A) $\frac{1}{6}$

(B) $\frac{1}{3}$

(C) $\frac{1}{2}$

(D) 6

[Ans.: (C)]

QUESTIONS ON LOG



DIBY-79

The value of $\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}} \times (\sqrt{7})^{\frac{1}{\log_4 7}}$ is _____.

[JEE (Adv.)-2018, Paper-1]

[Ans. : 8]

DIBY-80(A)

The value of $6 + \log_{\frac{3}{2}} \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \dots}}} \right)$ is _____.

[IIT-JEE-2012]

[Ans. : 04]

QUESTIONS ON LOG



DIBY-80(B)

The sum of squares of all the real solutions of the equation

$$\log_{(x+1)}(2x^2 + 5x + 3) = 4 - \log_{(2x+3)}(x^2 + 2x + 1) \text{ is equal to}$$

[JEE Mains-2026]

[Ans. : 2]

DIBY-81

The sum of all the real solutions of the equation

$$\log_{(x+3)}(6x^2 + 28x + 30) = 5 - 2\log_{(6x+10)}(x^2 + 6x + 9) \text{ is equal to}$$

[JEE Mains-2026]

(A) 1

(B) 2

(C) 0

(D) 4

[Ans. : (C)]

DIBY-82

The product of all solutions of the equation $e^{5(\log_e x)^2 + 3} = x^8, x > 0$, is:

[JEE Mains-2025]

(A) e^2

(B) e

(C) $e^{\frac{8}{5}}$

(D) $e^{\frac{6}{5}}$

[Ans. : (C)]

QUESTIONS ON LOG



DIBY-83

If for $x \in \left(0, \frac{\pi}{2}\right)$, $\log_{10} \sin x + \log_{10} \cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, $n > 0$, then the value of n is equal to: **[JEE Mains-2021]**

- (A) 16 (B) 12 (C) 9 (D) 20

[Ans.: (B)]

DIBY-84

The number of distinct solutions of the equation, $\log_{1/2} |\sin x| = 2 - \log_{1/2} |\cos x|$ in the interval $[0, 2\pi]$, is ____ **[JEE Mains-2020]**

[Ans.: 8]

DIBY-85

The sum of the roots of the equation,

$x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$, is: **[JEE Mains-2021]**

- (A) $\log_2 12$ (B) $\log_2 13$ (C) $\log_2 11$ (D) $\log_2 14$

[Ans.: (c)]

QUESTIONS ON MODULUS



DIBY-86

The number of real solution of the equation, $x^2 - |x| - 12 = 0$ is [JEE Mains-2021]

- (A) 2 (B) 3 (C) 1 (D) 4

[Ans. :]

DIBY-87

Solve $|x^2 + 4x + 3| + 2x + 5 = 0$

[JEE-1987, 5M]

[Ans. : $x = -4$ and $(-1 - \sqrt{3})$]

DIBY-88

The sum of all the real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is _____.

[JEE-1997, 2M]

[Ans. : 4]

QUESTIONS ON MODULUS



DIBY-89

$(x-3)^2 + |x-3| - 11 = 0$, the sum of solutions of this equation is?

- (A) 5 (B) 6 (C) 7 (D) 8

[Ans.: (B)]

DIBY-90

The sum of all roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is

[JEE Mains-2023]

- (A) $9 - \sqrt{3}$ (B) $9 + \sqrt{3}$ (C) $11 - \sqrt{3}$ (D) $11 + \sqrt{3}$

[Ans.: (B)]

DIBY-91

Exhaustive set of values of x satisfying $\log_{|x|}(x^2 + x + 1) \geq 0$ is

- (A) $(-1, 0)$ (B) $(-\infty, -1) \cup (1, \infty)$
(C) $(-\infty, \infty) \sim \{-1, 0, 1\}$ (D) $(-\infty, 1) \cup (-1, 0) \cup (1, \infty)$

[Ans.: (D)]

DIBY-92

Let $[t]$ denote the greatest integer $\leq t$. Then the equation in x ,

$$[x]^2 + 2[x + 2] - 7 = 0 \text{ has:}$$

[JEE Mains-2020]

- | | |
|-------------------------------|-------------------------------------|
| (A) Infinitely many solutions | (B) Exactly two solutions |
| (C) No integral solutio | (D) Exactly four integral solutions |

[Ans.: (A)]

DIBY-93

If $[x]$ be the greatest integer less than or equal to x , then $\sum_{n=8}^{100} \left[\frac{(-1)^n n}{2} \right]$ is equal to:

[JEE Mains-2021]

- | | | | |
|-------|--------|-------|-------|
| (A) 4 | (B) -2 | (C) 2 | (D) 0 |
|-------|--------|-------|-------|

[Ans.: (A)]

DIBY-94

Find the domain of the function $f(x) = \frac{1}{\sqrt{[x]^2 - 7[x] + 10}}$,

where $[.]$ denotes the greatest integer function.

[Ans. : $x \in (-\infty, 2) \cup [6, \infty)$]

DIBY-95

Find no. of solution of equation $x^2 - 12x + 35 = [x] + [-x]$ (where $[x]$ denotes the largest integer less than or equal to x).

[Ans. : 2]

DIBY-96

If $[x]$ denotes the integral part of x for real x , then the value of

$$\left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{200}\right] + \left[\frac{1}{4} + \frac{1}{100}\right] + \left[\frac{1}{4} + \frac{3}{200}\right] + \dots + \left[\frac{1}{4} + \frac{199}{200}\right]$$

[Ans. : 50]

QUESTIONS ON GIF

DIBY-97

Find the value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{2946}{1000}\right]$
where $[.]$ greatest integer function.

[Ans. : 4341]

Question [JEE Mains-2019] # DIBY-98

For $x \in R$, let $[x]$ denote the greatest integer $\leq x$, then the sum of the series

$$\left[-\frac{1}{3}\right] + \left[-\frac{1}{3} - \frac{1}{100}\right] + \left[-\frac{1}{3} - \frac{2}{100}\right] + \dots + \left[-\frac{1}{3} - \frac{99}{100}\right] \text{ is}$$

[Ans. : (c)]

A -135

B -153

C -133

D -131

QUESTIONS ON GIF & FPF



DIBY-99

If $2[x] + 3x = 4 - 5\{x\}$ where $[.]$ is GIF and $\{.\}$ is FPF. Find x .

[Ans. : (1/2)]

DIBY-100

Solve the following equation for x

$|x - 1| = 2[x] - 3\{x\}$ where $[x]$ and $\{x\}$ denotes integral and fractional part of x .

[Ans. : (3/2 & 11/4)]

DIBY-101

Let $[.]$ denote the greatest integer function. If the domain of the function

$f(x) = \cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to [JEE Mains-2026]

(A) 6

(B) 8

(C) 9

(D) 4

[Ans. : (A)]

DIBY-102

The domain of the function $f(x) = \sin^{-1}[2x^2 - 3] + \log_2 \left(\log_{\frac{1}{2}}(x^2 - 5x + 5) \right)$, where $[t]$ is the greatest integer function, is **[JEE Mains-2022]**

- (A) $\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2}\right)$ (B) $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$ (C) $\left(1, \frac{5-\sqrt{5}}{2}\right)$ (D) $\left[1, \frac{5+\sqrt{5}}{2}\right)$

[Ans.: (c)]

DIBY-103

If $S = \{a \in R: |2a - 1| = 3[a] + 2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t , then $72 \sum_{a \in S} a$ is **[JEE Mains-2024]**

[Ans.: 18]

DIBY-104

Total number of solutions of $[x]^2 = x + 2\{x\}$, where $[.]$ and $\{.\}$ denotes the greatest integer function and fractional part respectively, is equal to:

[Ans. : 4]

DIBY-105

If $[x^2] + x - a = 0$ has a solution where $a \in \mathbb{N}$ and $a \leq 20$, then total number of different values of ' a ' can be:

[Ans. : 4]

DIBY-106

Solve the following equation for x : $2x + 3[x] - 4\{-x\} = 4$
(where $[*]$ and $\{*\}$ denotes integral and fractional part of x)

[Ans. : 3/2]

DIBY-107

QUESTIONS ON GIF & FPF



Let $A = \{x \in \mathbb{R} : [x+3] + [x+4] \leq 3\}$, $B = \left\{x \in \mathbb{R} : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$, where $[t]$ denotes greatest integer function. Then, [JEE Mains-2023]

- (A) $A \cap B = \phi$ (B) $A = B$ (C) $B \subset A, A \neq B$ (D) $A \subset B, A \neq B$

[Ans.: (B)]

DIBY-108 [Challenger]

If $[.]$ denotes the greatest integer function and $\{.\}$ denotes the fractional part,

then the domain of the function $f(x) = \sqrt{\frac{x-1}{x-2\{x\}}} + \ln(3 - [x + [2x]])$ is given by

- (A) $(-\infty, 0) \cup (0, 1)$ (B) $\mathbb{R} - \{1\}$
(C) $(0, \frac{1}{2})$ (D) $(0, \infty)$

[Ans.: (A)]

DIBY-109

Let $[x]$ denote the greatest integer less than or equal to x . Then the values of $x \in R$ satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lies in the interval

[JEE Mains-2019]

- (A) $\left[0, \frac{1}{e}\right)$ (B) $[\log_e 2, \log_e 3]$ (C) $[1, e)$ (D) $[0, \log_e 2)$

[Ans.: (D)]

DIBY-110 (Challenger)

Let $[x]$ = the greatest integer less than or equal to x . If all the values of x such that the product $\left[x - \frac{1}{2}\right] \left[x + \frac{1}{2}\right]$ is prime, belongs to the set $[x_1, x_2) \cup [x_3, x_4)$, find the value of $x_1^2 + x_2^2 + x_3^2 + x_4^2$.

[Ans.: 11]

DIBY-111

Domain of $f(x) = \sqrt{2\{x\}^2 - 3\{x\} + 1}$ where $\{\cdot\}$ denotes the fractional part, in $[-1, 1]$ is

[Ans.: $x \in \left[-1, -\frac{1}{2}\right] \cup \left[0, \frac{1}{2}\right] \cup \{1\}$]

DIBY-112

Find domain of $f(x) = \frac{1}{[x]} + \log_{1-\{x\}}(x^2 - 3x + 10) + \frac{1}{\sqrt{2-|x|}} + \frac{1}{\sqrt{\sec(\sin x)}}$

where $[x]$ denotes the greatest integer function & $\{x\}$ denotes the fractional part function.

[Ans. : $(-2, -1) \cup (-1, 0) \cup (1, 2)$]

DIBY-113

Find the value of expression

$$\left[\frac{1022}{2023} \right] + \left[\frac{1022(2)}{2023} \right] + \left[\frac{1022(3)}{2023} \right] + \dots + \left[\frac{1022(2021)}{2023} \right] + \left[\frac{1022(2022)}{2023} \right].$$

[Note : $[y]$ denotes the greatest integer less than or equal to y]

[Ans. : 1021×1011]

RANGE OF FUNCTION

DIBY-114

Find Range of following functions:

- (i) $y = \cos^2 x + 6\cos x + 11$
- (ii) $y = 4^x - 2^x + 1$
- (iii) $f(x) = -5\cos^2 x - 8\sin x + 15$
- (iv) $f(x) = 3x^2 + 24|x| + 59$

Ans. :

- (i) $y \in [6, 18]$
- (ii) $y \in \left[\frac{3}{4}, \infty\right)$
- (iii) $y \in \left[\frac{34}{5}, 23\right]$
- (iv) $y \in [59, \infty)$

DIBY-115

The greatest value of expression $\log_{(14+4\sqrt{10})}(\cos^2 \theta - 6\sin \theta \cos \theta + 3\sin^2 \theta)$ is

[Ans. : 1/2]

DIBY-116

The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

[Ans. : 6]

RANGE OF FUNCTION



DIBY-117

Find the range of the following function : $f(x) = \frac{2e^x}{3e^x + 5}$

$$[\text{Ans.} : f(x) \in \left(0, \frac{2}{3}\right)]$$

DIBY-118

Range of $f(x) = \frac{\{x\}-2}{1-3\{x\}}$ (where $\{.\}$ represents FPF)

$$[\text{Ans.} : (-\infty, -2] \cup \left(\frac{1}{2}, \infty\right)]$$

DIBY-119

Find range of $f(x) = \frac{x^2}{1+x^2}$.

$$[\text{Ans.} : [0, 1)]$$

DIBY-120

Find the range of $f(x)$:

(i) $f(x) = 2x^2 - 3x + 2$

(ii) $f(x) = 2x^2 - 3x + 2, x \in [0, 2]$

(iii) $f(\theta) = 2\cos^2\theta - 6\sin\theta + 1$

$$\text{Ans.} : \left[\frac{7}{8}, \infty\right)$$

$$\text{Ans.} : \left[\frac{7}{8}, 4\right]$$

$$\text{Ans.} : [-5, 7]$$



RANGE OF FUNCTION

DIBY-121

Find the range of: $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}, x \in R$

[IITJEE-2003, 2M]

[Ans.: $(1, \frac{7}{3}]$]

DIBY-122

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \log_{\sqrt{m}}\{\sqrt{2}(\sin x - \cos x) + m - 2\}$, for some m , such that the range of f is $[0, 2]$. Then the value of m is [JEE Mains-2023]

- (A) 5 (B) 3 (C) 2 (D) 4

[Ans.: (A)]

DIBY-123

The range of the function:

$$f(x) = \log_{\sqrt{5}} \left(3 + \cos \left(\frac{3\pi}{4} + x \right) + \cos \left(\frac{\pi}{4} + x \right) + \cos \left(\frac{\pi}{4} - x \right) - \cos \left(\frac{3\pi}{4} - x \right) \right)$$

[JEE Mains-2021]

- (A) $[-2, 2]$ (B) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (C) $(0, \sqrt{5})$ (D) $[0, 2]$

[Ans.: (D)]

RANGE OF FUNCTION

DIBY-124

Match the following:

Column-I ($f(x)$)

(A) $f(x) = |x - 1| + |x - 2| + |x - 3|$

(B) $f(x) = \cos^2 x - 4\sin x + 7$

(C) $f(x) = \frac{7}{x^2 - 2x}$

(D) $f(x) = 4^x - 3 \cdot 2^{x+1} + 11$

Column-II (Range)

(P) $[3, 11]$

(Q) $[0, 1]$

(R) $[2, \infty)$

(S) $(-\infty, -7] \cup (0, \infty)$

[Ans. : A-R, B-P, C-S, D-R]

DIBY-125

Find the range of $f(x) = \cos^4 x + \sin^4 x$.

[Ans. : $y \in \left[\frac{1}{2}, 1\right]$]

RANGE OF FUNCTION

DIBY-126

Find range of $f(x) = \frac{|x|+4}{7-|x|}$.

- (A) $(-\infty, -1)$ (B) $\left[\frac{4}{7}, \infty\right)$ (C) $\left(-1, \frac{4}{7}\right]$ (D) $(-\infty, -1) \cup \left[\frac{4}{7}, \infty\right)$

[Ans.: (D)]

DIBY-127

Let $f(x) = \frac{1}{7-\sin 5x}$ defined on R . Range of $f(x)$ is:

[JEE Mains-2024]

- (A) $\left[\frac{1}{8}, \frac{1}{5}\right]$ (B) $\left[\frac{1}{7}, \frac{1}{6}\right]$ (C) $\left[\frac{1}{7}, \frac{1}{5}\right]$ (D) $\left[\frac{1}{8}, \frac{1}{6}\right]$

[Ans.: (D)]

DIBY-128

If the range of the function $f(x) = \frac{5-x}{x^2-3x+2}$, $x \neq 1, 2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to:

[JEE Mains-2025]

- (A) 190 (B) 192 (C) 188 (D) 194

[Ans.: (D)]

DIBY-129

Let $f: \mathbb{R} - \{2, 6\} \rightarrow \mathbb{R}$ be real valued function defined as $f(x) = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$. Then range of f is
[JEE Mains-2023]

(A) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(B) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(C) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

(D) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

[Ans.: (A)]

DIBY-130

Let the range of the function $f(x) = \frac{1}{2 + \sin 3x + \cos 3x}$, $x \in R$ be $[a, b]$

If α and β are respectively the A.M. and the G.M. of a and b , then $\frac{\alpha}{\beta}$ is equal to:

[JEE Mains-2024]

(A) $\sqrt{2}$

(B) 2

(C) $\sqrt{\pi}$

(D) π

[Ans.: (A)]

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians